

pathology

Part I: Multiple Choice Questions (Definitions)

Choose the most appropriate answer for each question:

1. Localized suppurative acute inflammation characterized by the formation of pus is called:
A) Carbuncle
B) Abscess
C) Furuncle
D) Cellulitis
2. Nuclear shrinkage characterized by dense chromatin and increased basophilia is:
A) Karyolysis
B) Karyorrhexis
C) Pyknosis
D) Necrosis
3. A small abscess related to a hair follicle or sebaceous gland is a:
A) Carbuncle
B) Furuncle (Boil)
C) Sinus
D) Fistula
4. The fragmentation of a pyknotic nucleus into multiple small fragments is known as:
A) Karyolysis
B) Karyorrhexis
C) Hyperplasia
D) Metaplasia
5. A **carbuncle** is best described as:
A) A single small abscess
B) Multiple communicating abscesses
C) A blind-ending tract
D) A benign tumor of fatty tissue
6. The pathological accumulation of water leading to cell swelling and cytoplasmic granularity is:
A) Fatty change

- B) Cloudy swelling
 - C) Necrosis
 - D) Oedema
7. The process of ingestion and destruction of bacteria and necrotic debris by inflammatory cells is:
- A) Chemotaxis
 - B) Phagocytosis
 - C) Opsonization
 - D) Embolization
8. The pathological accumulation of neutral fat in parenchymatous cells is called:
- A) Cloudy swelling
 - B) Fatty change
 - C) Necrosis
 - D) Lipoma
9. The directed movement of inflammatory cells toward an irritant in the area of inflammation is:
- A) Phagocytosis
 - B) Diapedesis
 - C) Chemotaxis
 - D) Margination
10. The abnormal accumulation of fluid in interstitial tissues and serous membranes is:
- A) Inflammation
 - B) Oedema
 - C) Thrombosis
 - D) Congestion
11. A chronic specific inflammation characterized by focal accumulation of modified macrophages (epithelioid cells) is a:
- A) Thrombus
 - B) Granuloma
 - C) Papilloma
 - D) Keloid
12. The formation of a compact mass from blood elements inside the cardiovascular system during life is:
- A) Embolism

B) Thrombosis

C) Ischemia

D) Infarction

13. Replacement of damaged cells by new ones of the same kind is defined as:

A) Scarring

B) Fibrosis

C) Regeneration

D) Hypertrophy

14. A compact mass formed of circulating blood elements during life is a:

A) Vegetation

B) Embolus

C) Thrombus

D) Clot

15. Cells that proliferate continuously throughout life to replace aging cells are:

A) Stable cells

B) Permanent cells

C) Labile cells

D) Fixed cells

16. A thrombus primarily composed of platelets and fibrin, occurring in the early phase, is a:

A) Red thrombus

B) Mixed thrombus

C) Pale thrombus

D) Septic thrombus

17. Cells that are unable to divide or regenerate, such as nerve cells, are:

A) Labile cells

B) Stable cells

C) Permanent cells

D) Epithelial cells

18. A thrombus that contains bacteria, often leading to pyemic abscesses, is a:

A) Pale thrombus

B) Aseptic thrombus

C) Septic thrombus

D) Mixed thrombus

19. Cells that do not normally divide at a significant rate but can do so when needed (e.g., liver) are:
- A) Labile cells
 - B) Stable cells**
 - C) Permanent cells
 - D) Muscle cells
20. A detached and circulating intravascular mass that impacts at a site distant from its origin is an:
- A) Thrombus
 - B) Embolus**
 - C) Ulcer
 - D) Vegetation
21. A local defect of the surface of an organ or tissue produced by the sloughing of inflammatory necrotic tissue is an:
- A) Ulcer**
 - B) Sinus
 - C) Fistula
 - D) Keloid
22. Thrombi formed specifically on heart valves are referred to as:
- A) Agonal thrombi
 - B) Mural thrombi
 - C) Vegetations**
 - D) Capillary thrombi
23. A protuberant scar covered by stretched epidermis due to overdone repair is a:
- A) Epidermal cyst
 - B) Keloid**
 - C) Fibroma
 - D) Stricture
24. A thrombus occurring in an inflamed vein (often due to bacteria) is:
- A) Phlebothrombosis
 - B) Thrombophlebitis**
 - C) Arteritis
 - D) Ischemia
25. Inflammation of a mucous membrane resulting in excess mucus (e.g., rhinitis) is:
- A) Serous inflammation

- B) Fibrinous inflammation
 - C) Catarrhal inflammation**
 - D) Allergic inflammation
26. A benign tumor originating from glandular epithelium is an:
- A) Papilloma
 - B) Adenoma**
 - C) Carcinoma
 - D) Leiomyoma
27. Inflammation characterized by excess watery fluid exudate (e.g., burn blisters) is:
- A) Serous inflammation**
 - B) Haemorrhagic inflammation
 - C) Catarrhal inflammation
 - D) Membranous inflammation
28. A benign tumor of fatty tissue is a:
- A) Fibroma
 - B) Lipoma**
 - C) Sarcoma
 - D) Chondroma
29. The local death of a large group of cells or tissues within a living body is:
- A) Apoptosis
 - B) Necrosis**
 - C) Ischemia
 - D) Putrefaction
30. A benign tumor of fibrous tissue is a:
- A) Fibroma**
 - B) Lipoma
 - C) Fibrosarcoma
 - D) Leiomyoma
31. The dissolution of the nucleus where it fails to take stain due to chromatin hydrolysis is:
- A) Pyknosis
 - B) Karyorrhexis
 - C) Karyolysis**
 - D) Atrophy

32. A benign tumor originating from cartilage is a:

- A) Osteoma
- B) Chondroma
- C) Fibroma
- D) Adenoma

33. A benign tumor of the protective surface epithelium is a:

- A) Papilloma
- B) Adenoma
- C) Carcinoma
- D) Metaplasia

34. A benign tumor originating from smooth muscle is a:

- A) Rhabdomyoma
- B) Leiomyoma
- C) Leiomyosarcoma
- D) Lipoma

35. The transformation of one differentiated cell type to another differentiated cell type is:

- A) Dysplasia
- B) Hyperplasia
- C) Metaplasia
- D) Neoplasia

36. A benign tumor of bone is an:

- A) Chondroma
- B) Osteoma
- C) Osteosarcoma
- D) Fibroma

37.

a) An increase in the number of cells leading to an increase in organ size is:

- A) Hypertrophy
- B) Hyperplasia
- C) Neoplasia
- D) Metaplasia

b) An increase in the size of individual cells leading to an increase in organ size is:

- A) Hypertrophy
- B) Hyperplasia

- C) Dysplasia
- D) Regeneration

38. Which test is considered a localized induration indicating a hypersensitivity reaction to tubercle bacilli?
- A) PCR
 - B) Sputum culture
 - C) Tuberculin test
 - D) Ziehl–Neelsen stain
39. The black color in gangrene is specifically due to the reaction of hydrogen sulphide with:
- A) Calcium
 - B) Iron of hemoglobin (forming iron sulphide)
 - C) Carbon dioxide
 - D) Necrotic protein
40. Barrett's esophagus is a clinical example of metaplasia where squamous epithelium changes into:
- A) Squamous cell carcinoma
 - B) Intestinal (columnar) epithelium
 - C) Transitional epithelium
 - D) Fibrous tissue
41. A tumor that is locally invasive and destructive but does not cause distant metastasis is classified as:
- A) Benign
 - B) Malignant
 - C) Locally malignant (Intermediate)
 - D) Carcinoma in situ
42. Which of the following is a compensatory type of hypertrophy?
- A) Uterus in pregnancy
 - B) Heart in hypertension
 - C) Kidney enlargement after removal of the other kidney
 - D) Muscle growth in athletes
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Part II: Compare Between

1. Acute and Chronic Inflammation

Feature	Acute Inflammation	Chronic Inflammation
Onset	Fast (minutes to hours)	Slow (days)
Cellular infiltrate	Mainly neutrophils	Monocytes/macrophages and lymphocytes
Tissue injury	Usually mild and self-limited	Often severe and progressive
Signs	Local and systemic signs are prominent	Less prominent

2. Primary and Secondary Tuberculosis

Feature	Primary Tuberculosis	Secondary Tuberculosis
Exposure	First-time infection (usually children)	Reinfection or reactivation (usually adults)
Organs	Lung, intestine, tonsil	Can affect any organ
Lymphadenitis	Prominent	Minimal
Tissue destruction	Less marked	More marked (hypersensitivity)

3. Healing by Primary and Secondary Intention

Feature	Primary Intention	Secondary Intention
Cleanness	Clean	Unclean
Margins	Surgically clean margins	Irregular margins
Healing period	Short	Long
Outcome	Neat linear scar	Contracted irregular wound

Part III: Enumeration

1. Causes of Inflammation

- **Living irritants** (bacteria, viruses, parasites, fungi)
 - **Physical irritants** (excess heat, cold, radiation)
 - **Chemical irritants** (acids, alkalis, poisons)
 - **Mechanical irritants** (trauma, friction)
 - **Immunological** (allergic inflammation)
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2. Factors Affecting Repair

Local factors:

- Wound type and size
- Movement
- Infection
- Blood supply
- Foreign material
- Radiation

General factors:

- Age
 - General health (e.g., diabetes)
 - Drugs (e.g., steroids)
 - Cardiovascular status
 - Dietary deficiencies (protein, vitamin C)
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3. Complications of Repair

- **Insufficient fibrosis** (ulcer, sinus, fistula)
- **Excessive fibrosis** (keloid)
- **Excessive contraction** (contractures, strictures)

- **Epidermal cyst** (implantation cyst)
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4. Types of Gangrene

- Dry gangrene
 - Moist (wet) gangrene
 - Infective gangrene
(gas gangrene, lung gangrene, bed sores, cancrum oris)
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5. Fates of Thrombus

- Lysis
 - Organization
 - Recanalization
 - Embolization
 - Calcification
 - Occlusion
 - Infection
-

6. Routes of Spread of Malignant Tumours

- Direct spread (invasion)
 - Lymphatic spread
 - Vascular (blood) spread
 - Transcoelomic spread
 - Perineural spread
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7. Components of Granuloma

- Epithelioid cells (modified macrophages)

- Giant cells (Langhans or foreign body type)
 - Lymphocytes
 - Fibroblasts
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Part V: Essential Enumerations

Evidence of Malignant Transformation

1. Increased rate of growth
2. Loss of cellular differentiation
3. Infiltration of capsule and surrounding tissue

Fate of a Septic Thrombus

- Fragmentation → septic emboli → pyemic abscesses

Mechanism of Edema in Heart Failure

- Increased capillary hydrostatic pressure
- Salt and water retention (aldosterone)
- Increased permeability due to hypoxia

Labile cells:

- Proliferate continuously throughout life to replace aging cells
- Usually rapid regeneration
- They include → Epithelium, Lymphocytes

Stable cells:

Not normally dividing at a significant rate, only on need.

- Speed of regeneration variable
- Small damage → regeneration.
- Large damage → repair with fibrosis
- Includes → Chondrocytes, fibroblasts, liver cells, renal cells

Permanent cells:

- Unable to divide
- Unable to regenerate
- Include → Cardiac muscle cells, Nerve cells